ADNESH DHAMANGAONKAR

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EDUCATION

Northeastern University, Boston, USA — **Sep 2025 - May 2027**
Master of Science in Computer Science — **GPA: 4/4**
College of Engineering, Pune, India — **Aug 2017 - May 2021**
Bachelor of Technology in Computer Engineering — **CGPA: 8.27/10**

TECHNICAL SKILLS

Programming Languages	Java, Python, C, C++, JavaScript, Typescript
Libraries/Frameworks	Spring Boot, Spring Batch, Flask, React Native, Expo
Web Technologies	HTML, CSS, Angular, React, Node.js, Express
Tools/Databases	Git, Jenkins, Splunk, Dynatrace, AWS, PCF, Chef, MySQL, MongoDB, DB2, Oracle
Other	Claude Code, Cursor, Agile development, Linux, Bash, IntelliJ, Eclipse

WORK EXPERIENCE

SQA Engineer Coop | Nokia | Sunnyvale, CA, USA — **May 2026 - Current**

- Working on pyshl, Nokia's large-scale Python (**pytest**) test automation framework for SR Linux carrier-grade routers, gaining deep familiarity with a production-scale codebase.
- Enhanced a **Cursor skill** for automated test execution on local Docker DUT topologies by implementing multi-setup detection, enabling the skill to identify and select the correct Docker environment when multiple setups with varying topologies are present.
- Added topology and sub-topology validation to the skill prior to test execution, preventing runs on mismatched configurations and eliminating redundant LLM-driven failure investigations, saving both engineering time and LLM token consumption.
- Developed a new Cursor skill to execute pyshl test cases on remote testbeds, mirroring the local Docker workflow with automated code sync, execution, and post-run analysis, printing a summary on pass and performing failure investigation by ssh logging in the testbed on failure.
- Designed and implemented MT test suite for subinterface existence detection feature in SR Linux Vlan detection module in pyshl.

Software Engineer II | Mastercard | Pune, India — **Mar 2023 - Aug 2025**

- Utilized Mastercard's event framework to create a reporting application which generates events according to database changes.
- Modernized legacy **Java Spring Batch** modules to move them out of mainframe to run on cloud and VM platform.
- Optimized and increased performance of spend qualification batch modules **by 50%** by utilizing parallel processing with threads.
- Created a **Spring Boot** application to create REST APIs related to encryption, decryption, hash for utilizing CaaS (Cryptography as a Service) and achieve performance of **4500 tps** for the APIs.
- Owned the application lifecycle from development to deployment in production using **CI/CD pipelines with Jenkins.**

Software Engineer I | Mastercard | Pune, India — **Jul 2021 - Feb 2023**

- Created a **Spring Boot** application from scratch which exposes APIs related to file uploads, downloads, search etc. to handle customer files and send report files to customers and oversaw the lifecycle of application from development to production.
- Optimized the code to improve the performance of file uploads and downloads and create chunk upload and download APIs which **reduced file transfer failures** significantly.
- Created a dashboard for the batch jobs and LRMs real time status with **Angular** in frontend and **Spring Boot** in backend.
- Optimized SQL queries of **Spring Batch** modules to improve the batch performance **by 30%.**

Intern | Mastercard | Pune, India — **May 2020 - Jun 2020**

- Worked with **Python RASA framework** and **Angular** framework to design a chatbot for Mastercard website.
- Focused on intent classification in chatbot.
- Deployed the chatbot on Mastercard Careers page and interacts with users to help them navigate the website.

PUBLICATIONS

Secure, Decentralized, Privacy Preserving Machine Learning Implementation over Blockchain
https://doi.org/10.1145/3475992.3476003
3rd Blockchain and Internet of Things Conference, Vietnam (BIOTC - 2021)

- Authored research paper in **BIOTC 2021** supported by ACM Singapore Chapter, indexed by Ei Compendex, submitted to Scopus, presented the same at conference and received "**Best Presentation Award**".
- Proposed novel solution for implementing a decentralized machine learning model over a blockchain where training data can be submitted by multiple participating parties.
- Implemented privacy of individual data by utilizing gradient descent algorithm and differential privacy.
- Proposed a "**Within-N**" algorithm to discard false data vectors sent by malicious nodes, thus improving reliability.
- Implemented the proposed model as a proof of concept with React, Flask, Ganache blockchain and Solidity language.

ACADEMIC PROJECTS

Games - Created games like Chain Reaction Game, Nokia Snake Game Replica for PC, Snake vs Block Game using SDL2 library.

AI with Maze - Implementations of various artificial intelligence algorithms for solving the maze problem in Python; useful for learners who want to learn and compare artificial intelligence algorithms like A-star, BFS, DFS, etc. Visualized the flow of algorithms and plotted graphs of time taken to compare the performance.

ACHIEVEMENTS

- Received "**Expectations Exceeded**" performance rating for years 2022, 2023 and 2024 in Mastercard.
- Finalist in **Smart India Hackathon** (implemented online judiciary system with video conferencing feature).
- **Second prize** in national level Geek-O-Code coding competition.
- **Best Presentation Award** at BIOTC 2021.